Efficiency of coordinate descent methods on huge-scale optimization problems – Final Project

Kevin Wu

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1 Introduction

Large-scale optimization problems have attracted growing attention as datasets become both larger and more complex. In machine learning and computational mathematics, the time complexity of evaluating the objective function and its gradient is often a central concern. When the data dimension is high, computing the full gradient can be prohibitively expensive. This naturally motivates computational cheaper alternatives that use only partial gradients and move along individual directions to reduce the objective. Coordinate descent methods are one such class of algorithms: they iteratively minimize the function along coordinate directions or blocks of coordinates, choosing a coordinate or block at each step according to some rule, and then optimizing over the corresponding coordinate subspace while keeping the others fixed. Nevertheless, standard coordinate descent methods can be subject to criticism in several domains:

- **Theoratical Aspect:** One of the easiest variant of this class of method is *Cyclic coordinate search*. However, it is subtle or impossible to prove the rate of convergence[1)]. Intuitively, with fixed and cyclic order, the method cannot have a clean "potential decrease per step". Another method is to decrease along the direction with the maximal gradient, choose $i_k = arg \max_{1\leq i\leq n} |\nabla_i f(x_k)|$ and then update $x_{k+1} = x_k - \frac{1}{M}\nabla_{i_k} f(x_k) e_{i_k}$. However, this method also need to compute the full gradient. Thus, there is no computational advantage and the convergence rate is slightly slower than the naive gradient descent.
- **Time Complexity:** Line search is the most common strategy while implementing classic coordinate descent. Based on automatic differentiation, the time complexity of computing full gradient is proportional to the time complexity of computing the objective function value. Hence, for coordinate descent methods, the line-search strategies based on the objective function values are too example in practice. And intuitively, it is quite subtle to provide the theoratical guarantee for the global convergence based on this strategy.

Therefore, finding an algorithm based on coordinate descent methods that has computational advantage and theoretical guarantee is crucial. To survive, Nesterov came up with this Random Coordinate Descent Method.

2 Structure

This report is organized as follows. We first introduce the notation used throughout the paper. We then present the Random Coordinate Descent Method (RCDM) and analyze its convergence over

general convex functions and strongly convex functions. Finally, we extend RCDM to constrained optimization problems and discuss the corresponding convergence results.

3 Notation

The decomposition of $\mathbb{R}^N$ on n subspaces is written

$$\mathbb{R}^N = \bigotimes_{i=1}^{n} \mathbb{R}^{n_i}, \quad N = \sum_{i=1}^{n} n_i.$$

The corresponding partition of the unit matrix I_N is given by

$$I_N = (U_1, \cdots, U_n) \in \mathbb{R}^{N\times N}, \quad U_i \in \mathbb{R}^{N\times n},\ i = 1, \cdots, n.$$

For any $x = (x^{(1)}, \cdots, x^{(n)})^T \in \mathbb{R}^N$ can be decomposed as

$$x = \sum_{i=1}^{n} U_i x^{(i)}, \quad x^{(i)} \in \mathbb{R}^{n_i},\ i = 1, \cdots, n.$$

The partial gradient of $f(x)$ in $x^{(i)}$ is defined as

$$f_i'(x) = U_i^T \nabla f(x) \in \mathbb{R}^{n_i}, \quad x \in \mathbb{R}^N$$

4 Random Coordinate Descent Method (RCDM)

We consider the unconstrained minimization problem

$$\min_{x\in\mathbb{R}^N} f(x), \tag{4.1}$$

where f is convex and differentiable on $\mathbb{R}^N$, and the optimal set

$$X^\star := \arg\min_x f(x)$$

is nonempty. We denote $f^\star := \min_x f(x)$.

4.1 Coordinate-wise Lipschitz continuity

On each subspace (block) $\mathbb{R}^{n_i}$ we define primal norm $\|\cdot\|_{(i)}$ and its dual norm $\|\cdot\|_{(i)}^*$.

The key smoothness assumption is *coordinate-wise Lipschitz continuity*: for each block i there exists a Lipchitz constant $L_i > 0$ such that

$$\|f_i'(x + U_i h_i) - f_i'(x)\|_{(i)}^* \ \le\ L_i \|h_i\|_{(i)}, \quad \forall x \in \mathbb{R}^N,\ \forall h_i \in \mathbb{R}^{n_i},\ i = 1, \ldots, n. \tag{4.2}$$

Intuitively, we assume the function is strongly smooth.

From (4.2), we obtains a *quadratic upper bound* of block i:

$$f(x + U_i h_i) \ \le\ f(x) + \langle f_i'(x), h_i\rangle + \frac{L_i}{2}\|h_i\|_{(i)}^2, \quad \forall x \in \mathbb{R}^N,\ \forall h_i \in \mathbb{R}^{n_i}. \tag{4.3}$$

4.2 Weighted norms and global Lipschitz constant

Following Nesterov, we introduce a family of weighted norms parameterized by $\alpha \in \mathbb{R}$. For $x \in \mathbb{R}^N$ with blocks $x^{(i)}$, define

$$\|x\|_\alpha^2 := \sum_{i=1}^n L_i^\alpha \, \|x^{(i)}\|_{(i)}^2.$$

For a dual vector $g \in \mathbb{R}^N$ with blocks $g^{(i)}$, define the dual norm

$$\|g\|_\alpha^{*2} := \sum_{i=1}^n L_i^{-\alpha} \, \|g^{(i)}\|_{(i)}^{*2}.$$

We also define $S_\alpha(f)$ as:

$$S_\alpha(f) := \sum_{i=1}^n L_i^\alpha.$$

The coordinate-wise Lipschitz assumption (4.2) implies a global Lipschitz bound on the gradient with respect to the norm $\|\cdot\|_{1-\alpha}$.

Lemma 1 (Global Lipschitz gradient)**.** *For all $x, y \in \mathbb{R}^N$,*

$$\|\nabla f(x) - \nabla f(y)\|_{1-\alpha}^* \;\le\; S_\alpha(f) \, \|x - y\|_{1-\alpha}. \tag{4.4}$$

As a standard consequence of (4.4), we obtain a *quadratic upper bound* on f in the global norm:

$$f(x) \;\le\; f(y) + \langle \nabla f(y), x - y \rangle + \frac{S_\alpha(f)}{2} \, \|x - y\|_{1-\alpha}^2, \quad \forall x, y \in \mathbb{R}^N. \tag{4.5}$$

We also need a measure of the distance. Let

$$X^\star := \arg\min_x f(x), \qquad f^\star := \min_x f(x).$$

For $\beta \in \mathbb{R}$, define the distance between the feasible set and the optimal solution as:

$$R_\beta(x_0) := \max\Big\{ \max_{x^\star \in X^\star} \|x - x^\star\|_\beta : f(x) \le f(x_0) \Big\}.$$

4.3 Optimal coordinate steps and block decrease

Fix a block i. We define the *optimal block update* $T_i(x)$ as:

$$T_i(x) := x - \frac{1}{L_i} U_i f_i'(x) \tag{4.6}$$

Lemma 2. *For every $x \in \mathbb{R}^N$ and every block i,*

$$f(x) - f(T_i(x)) \;\ge\; \frac{1}{2L_i} \Big(\|f_i'(x)\|_{(i)}^* \Big)^2. \tag{4.7}$$

Proof. Plugging $T_i(x) = x - \frac{1}{L_i} U_i f_i'(x)$ into (4.3) to substitute $x + U_i h_i$

$$\begin{aligned} f(x) - f(x - \frac{1}{L_i} U_i f_i'(x)) &\ge \frac{1}{L_i} \Big(\|f_i'(x)\|^* \Big)^2 - \frac{1}{2L_i} \Big(\|f_i'(x)\|^* \Big)^2 \\ &\ge \frac{1}{2L_i} \Big(\|f_i'(x)\|_{(i)}^* \Big)^2. \end{aligned}$$

□

Updating block i yields at least $\frac{1}{2L_i} \|f_i'(x)\|_{(i)}^{*2}$ decrease in objective function value.

4.4 The RCDM(α, x_0) algorithm

We now describe the *Random Coordinate Descent Method* with parameter α and initial point x_0.

- **Input:** $x_0 \in \mathbb{R}^N$, exponent $\alpha \in \mathbb{R}$, block Lipschitz constants L_i.
- **Sampling distribution:** define

$$p_i^{(\alpha)} := \mathbb{P}\{i_k = i\} = \frac{L_i^\alpha}{S_\alpha(f)}, \quad i = 1, \dots, n.$$

- **Iteration:** for $k = 0, 1, 2, \dots$
  1. Sample a block index i_k according to the distribution $p^{(\alpha)}$.
  2. Update only block i_k:

$$x_{k+1} := T_{i_k}(x_k).$$

Let $\xi_k = (i_0, \dots, i_{k-1})$ denote the random sequence of chosen blocks up to iteration k, and define

$$\varphi_k := \mathbb{E}_{\xi_k}[f(x_k)]$$

as the expected objective function value at iteration k.

4.5 Gradient norm vs. suboptimality

We first relate the norm of the gradient to the function suboptimality. From Lemma 2 and the definition of S_α, one can show that for every x,

$$f(x) - f^\star \;\geq\; \frac{1}{2S_\alpha(f)} \|\nabla f(x)\|_{1-\alpha}^{*2}. \tag{4.8}$$

On the other hand, convexity implies for any $x \in \mathbb{R}^N$ and any $x^\star \in X^\star$,

$$f(x) - f^\star \;\leq\; \langle \nabla f(x), x - x^\star \rangle \;\leq\; \|\nabla f(x)\|_{1-\alpha}^{*} \, \|x - x^\star\|_{1-\alpha}. \tag{4.9}$$

Taking the best $x^\star \in X^\star$ and recalling the definition of $R_{1-\alpha}(x_0)$, we obtain for all x in the initial level set $\{f(x) \leq f(x_0)\}$:

$$f(x) - f^\star \;\leq\; R_{1-\alpha}(x_0) \, \|\nabla f(x)\|_{1-\alpha}^{*}. \tag{4.10}$$

Combining (4.8) and (4.10), we get useful inequalities for the gradient norm in terms of suboptimality:

$$\frac{1}{2S_\alpha(f)} \|\nabla f(x)\|_{1-\alpha}^{*2} \;\leq\; f(x) - f^\star \;\leq\; R_{1-\alpha}(x_0) \, \|\nabla f(x)\|_{1-\alpha}^{*}. \tag{4.11}$$

In particular, from the right inequality we can get

$$\|\nabla f(x)\|_{1-\alpha}^{*2} \;\geq\; \frac{(f(x) - f^\star)^2}{R_{1-\alpha}^2(x_0)}. \tag{4.12}$$

4.6 Expected decrease in one RCDM step

Using the block decrease (4.7), for any block i,

$$f(x_k) - f(T_i(x_k)) \;\geq\; \frac{1}{2L_i}\left(\|f_i'(x_k)\|_{(i)}^*\right)^2. \tag{4.13}$$

Taking expectation with respect to the random choice i_k with probabilities $p_i^{(\alpha)}$, we have

$$\mathbb{E}\big[f(x_k) - f(x_{k+1}) \mid x_k\big] = \sum_{i=1}^n p_i^{(\alpha)}\left(f(x_k) - f(T_i(x_k))\right) \tag{4.14}$$

$$\geq \sum_{i=1}^n p_i^{(\alpha)} \frac{1}{2L_i}\left(\|f_i'(x_k)\|_{(i)}^*\right)^2. \tag{4.15}$$

Plugging $p_i^{(\alpha)} = L_i^\alpha / S_\alpha(f)$ gives

$$\mathbb{E}\big[f(x_k) - f(x_{k+1}) \mid x_k\big] \;\geq\; \frac{1}{2S_\alpha(f)} \sum_{i=1}^n L_i^{\alpha-1} \|f_i'(x_k)\|_{(i)}^{*2} = \frac{1}{2S_\alpha(f)} \|\nabla f(x_k)\|_{1-\alpha}^{*2}. \tag{4.16}$$

Therefore,

$$\mathbb{E}\big[f(x_{k+1}) \mid x_k\big] \;\leq\; f(x_k) - \frac{1}{2S_\alpha(f)} \|\nabla f(x_k)\|_{1-\alpha}^{*2}. \tag{4.17}$$

Using (4.12) to lower-bound the gradient norm, we obtain for all x_k with $f(x_k) \leq f(x_0)$,

$$\mathbb{E}\big[f(x_{k+1}) - f^\star \mid x_k\big] \;\leq\; f(x_k) - f^\star - \frac{1}{2S_\alpha(f)\, R_{1-\alpha}^2(x_0)} \left(f(x_k) - f^\star\right)^2. \tag{4.18}$$

Define $\delta_k := \mathbb{E}[f(x_k)] - f^\star = \varphi_k - f^\star$. Taking expectation on (4.18), given:

$$\delta_{k+1} \;\leq\; \delta_k - c\,\delta_k^2, \qquad c := \frac{1}{2S_\alpha(f) R_{1-\alpha}^2(x_0)}. \tag{4.19}$$

This recurrence yields an $O(1/k)$ decay.

4.7 Convergence rate: proof of Theorem 1

We now state and prove the main convergence result for RCDM over general convex functions.

Theorem 1 (Convergence rate of RCDM)**.** *Assume f is convex, differentiable, and satisfies the coordinate-wise Lipschitz condition* (4.2)*. Let x_k be the iterates generated by RCDM(α, x_0). Then for all $k \geq 0$,*

$$\mathbb{E}_{\xi_k}[f(x_k)] - f^\star \;\leq\; \frac{2\, S_\alpha(f)}{k+4}\, R_{1-\alpha}^2(x_0). \tag{4.20}$$

Proof. We start from the recursion (4.19):

$$\delta_{k+1} \leq \delta_k - c\,\delta_k^2, \quad c > 0. \tag{4.21}$$

For any k such that $\delta_k > 0$, we have

$$\delta_{k+1} \leq \delta_k(1 - c\,\delta_k). \tag{4.22}$$

Using the inequality $\frac{1}{1-u} \geq 1+u$ for $u \in (0,1)$, we get

$$\frac{1}{\delta_{k+1}} \;\geq\; \frac{1}{\delta_k(1-c\,\delta_k)} \;\geq\; \frac{1}{\delta_k}(1+c\,\delta_k) = \frac{1}{\delta_k} + c. \tag{4.23}$$

Thus

$$\frac{1}{\delta_{k+1}} \;\geq\; \frac{1}{\delta_k} + c \;\geq\; \frac{1}{\delta_0} + (k+1)c. \tag{4.24}$$

Assigning k to $k+1$, given

$$\frac{1}{\delta_k} \;\geq\; \frac{1}{\delta_0} + kc, \tag{4.25}$$

hence

$$\delta_k \;\leq\; \frac{1}{\frac{1}{\delta_0} + kc}. \tag{4.26}$$

Using the bound $\delta_0 = f(x_0) - f^\star \leq S_\alpha(f)\, R^2_{1-\alpha}(x_0)$, we have

$$\frac{1}{\delta_0} \;\geq\; \frac{1}{S_\alpha(f)\, R^2_{1-\alpha}(x_0)}. \tag{4.27}$$

Combining this with $c = \frac{1}{2S_\alpha(f)R^2_{1-\alpha}(x_0)}$, we obtain the simpler inequality

$$\delta_k \;\leq\; \frac{2\, S_\alpha(f)\, R^2_{1-\alpha}(x_0)}{k+4}, \tag{4.28}$$

for all $k \geq 0$. This proves the claimed bound (4.20). □

4.8 Comments and special cases

Theorem 1 shows that RCDM achieves an $O(1/k)$ convergence rate in expectation, similar to the classical gradient method, but with much cheaper iterations.

Several special choices of α are particularly interesting:

- **Uniform sampling ($\alpha = 0$).** Then $S_0(f) = \sum_i L_i^0 = n$, and the bound becomes

  $$\mathbb{E}[f(x_k)] - f^\star \;\leq\; \frac{2n}{k+4}\, R_1^2(x_0).$$

  Compare to the full gradient descent:

  $$f(x_k) - f^* \leq \frac{n}{k} R_1^2$$

  Roughly speaking, random coordinate descent has a rate comparable to full-gradient descent, but each step is much cheaper.

- **Lipschitz-weighted sampling ($\alpha = 1$).** Here $S_1(f) = \sum_i L_i$, and we obtain

  $$\mathbb{E}[f(x_k)] - f^\star \;\leq\; \frac{2}{k+4}\Big(\sum_{i=1}^n L_i\Big) R_0^2(x_0).$$

  Compare to the full gradient descent:

  $$f(x_k) - f^* \leq \frac{\max_i\{L_i\}}{k} R_1^2$$

  Still, the convergence rate of random coordinate descent is proportional to one of full-gradient method.

- **Intermediate choice ($\alpha = \frac{1}{2}$).** Define

$$D_\infty(x_0) := \max_x \{ \max_{y \in X^*} \max_{1 \le i \le N} |x^{(i)} - y^{(i)}| : f(x) < f(x_0) \}$$

Then one can show

$$\mathbb{E}[f(x_k) - f^*] \le \frac{2}{k+4} \Big(\sum_{j=1}^n L_j^{1/2} \Big) D_\infty^2(x_0)$$

However, there exists some cases that the random coordinate descent method can work while the full-gradient method cannot in the sense that $\sum_{j=1}^n L_j^{1/2}$ can be bounded when $L = \max\{L_i\}$ cannot.

Overall, RCDM provides a theoretically well-justified and computationally cheap method for huge-scale convex optimization. It matches the $O(1/k)$ rate of full gradient methods while allowing each iteration to work with only one coordinate block, thus obtaining computational advantage.

5 RCDM over strongly convex functions

In this section we sfocus on the case where the objective function is not only convex, but *strongly convex*.

Recall the family of weighted norms

$$\|x\|_\alpha^2 \;=\; \sum_{i=1}^n L_i^\alpha \, \|x^{(i)}\|_{(i)}^2, \qquad \|g\|_\alpha^{*2} \;=\; \sum_{i=1}^n L_i^{-\alpha} \, \|g^{(i)}\|_{(i)}^{*2},$$

and

$$S_\alpha(f) := \sum_{i=1}^n L_i^\alpha.$$

5.1 Strong convexity assumption

We say that f is *strongly convex with respect to the norm* $\|\cdot\|_{1-\alpha}$ with parameter $\sigma_{1-\alpha}(f) > 0$ if

$$f(y) \;\ge\; f(x) + \langle \nabla f(x), y - x \rangle + \frac{\sigma_{1-\alpha}(f)}{2} \, \|y - x\|_{1-\alpha}^2, \qquad \forall x, y \in \mathbb{R}^N. \tag{5.1}$$

From (5.1), by minimizing the right-hand side in y, we obtain the standard gradient sub-optimality inequality:

$$f(x) - f^\star \;\le\; \frac{1}{2\,\sigma_{1-\alpha}(f)} \, \|\nabla f(x)\|_{1-\alpha}^{*2}, \qquad \forall x. \tag{5.2}$$

5.2 Linear convergence in expectation

Let $\{x_k\}$ be the iterates generated by RCDM(α, x_0) as defined in Section 4 of this report, and let

$$\varphi_k := \mathbb{E}_{\xi_{k-1}}[f(x_k)]$$

denote the expected objective value after k iterations.

Theorem 2 (RCDM on strongly convex functions)**.** *Suppose f is strongly convex with respect to $\|\cdot\|_{1-\alpha}$ with parameter $\sigma_{1-\alpha}(f) > 0$, and coordinate-wise Lipschitz continuous with constants L_i as before. Then the iterates generated by* RCDM(α, x_0) *satisfy*

$$\varphi_k - f^\star \;\le\; \left(1 - \frac{\sigma_{1-\alpha}(f)}{S_\alpha(f)}\right)^k (f(x_0) - f^\star), \qquad k \ge 0. \tag{5.3}$$

Thus, in the strongly convex case, RCDM enjoys a *linear* convergence rate in expectation, with factor

$$\rho_\alpha \;=\; 1 - \frac{\sigma_{1-\alpha}(f)}{S_\alpha(f)} \in (0,1).$$

Equivalently, the number of iterations needed to reduce the expected sub-optimality below a tolerance $\varepsilon > 0$ is bounded by

$$k \;\ge \frac{S_\alpha(f)}{\sigma_{1-\alpha}(f)} \log\!\left(\frac{f(x_0) - f^\star}{\varepsilon}\right),$$

The term

$$\kappa_\alpha := \frac{S_\alpha(f)}{\sigma_{1-\alpha}(f)}$$

can be seen as *condition number* for RCDM, analogous to L/μ for full gradient descent.

5.3 Comments and special cases

Several particular values of α are worth highlighting:

- **Uniform sampling ($\alpha = 0$).** In this case $S_0(f) = n$, so the rate becomes

  $$\varphi_k - f^\star \;\le\; \left(1 - \frac{\sigma_1(f)}{n}\right)^k (f(x_0) - f^\star).$$

  The convergence rate depends on n (the number of blocks) and on the strong convexity parameter with respect to $\|\cdot\|_1$.

- **Lipschitz-weighted sampling ($\alpha = 1$).** Here $S_1(f) = \sum_{i=1}^n L_i$, and the rate becomes

  $$\varphi_k - f^\star \;\le\; \left(1 - \frac{\sigma_0(f)}{\sum_{i=1}^n L_i}\right)^k (f(x_0) - f^\star).$$

  In Euclidean norms this shows that the convergence rate depends on the ratio of an average curvature term $\sum_i L_i$ to the strong convexity parameter $\sigma_0(f)$, rather than on the global Lipschitz constant of the full gradient.

- **Intermediate choice ($\alpha = \frac{1}{2}$).** This yields an intermediate condition number $\kappa_{1/2} = S_{1/2}(f)/\sigma_{1/2}(f)$, which may be much smaller than the usual L/μ for certain classes of problems.

6 RCDM over constrained problems

In many applications, the decision variable is constrained to lie in a simple convex set. In this section we extend the random coordinate descent method to the constrained problem

$$\min_{x \in Q} f(x), \tag{6.1}$$

where the feasible set is a Cartesian product

$$Q = \bigotimes_{i=1}^{n} Q_i, \qquad Q_i \subset \mathbb{R}^{n_i} \text{ closed and convex.}$$

6.1 Constrained coordinate update

Given a current point $x \in Q$, we define the *constrained* update on block i by solving a local quadratic model with projection onto Q_i:

$$u^{(i)}(x) \;\in\; \arg\min_{u^{(i)} \in Q_i} \left\{ \langle f_i'(x),\, u^{(i)} - x^{(i)} \rangle + \frac{L_i}{2} \|u^{(i)} - x^{(i)}\|_{(i)}^2 \right\}, \tag{6.2}$$

and then lifting this block change back to $\mathbb{R}^N$:

$$V_i(x) \;=\; x + U_i\big(u^{(i)}(x) - x^{(i)}\big), \qquad i = 1, \dots, n. \tag{6.3}$$

By construction we have $u^{(i)}(x) \in Q_i$ and thus $V_i(x) \in Q$.

The optimality condition for (6.2) is

$$\big\langle f_i'(x) + L_i B_i\big(u^{(i)}(x) - x^{(i)}\big),\, u^{(i)} - u^{(i)}(x) \big\rangle \;\geq\; 0 \quad \forall\, u^{(i)} \in Q_i,\; i = 1, \dots, n. \tag{6.4}$$

Combining the quadratic upper bound (2.3) with (6.4) and choosing $u^{(i)} = x^{(i)}$ yields the key decrease estimate:

$$f(x) - f(V_i(x)) \;\geq\; \frac{L_i}{2} \|u^{(i)}(x) - x^{(i)}\|_{(i)}^2, \qquad i = 1, \dots, n. \tag{6.5}$$

6.2 Uniform Coordinate Descent Method (UCDM)

For the constrained problem Nesterov considers the *uniform* random choice of coordinates, i.e., $p_i = 1/n$ for all i. The algorithm is:

Method UCDM(x_0)
Input: starting point $x_0 \in Q$.
For $k \geq 0$:

1. Choose i_k uniformly from $\{1, \dots, n\}$.
2. Set $x_{k+1} = V_{i_k}(x_k)$, where V_i is defined in (6.3).

As before, write $\xi_k = \{i_0, \dots, i_{k-1}\}$ for the history of chosen coordinates and

$$\phi_k \;\equiv\; \mathbb{E}_{\xi_{k-1}} f(x_k)$$

for the expected objective value at iteration k.

6.3 Convergence rate for the constrained case

We first analyze the convex case, then the strongly convex case.

Convex case

Fix an optimal solution $x^* \in X^*$. Define the distance as

$$r_k^2 := \|x_k - x^*\|_1^2 = \sum_{i=1}^{n} L_i \left\langle B_i(x_k^{(i)} - x^{*(i)}), x_k^{(i)} - x^{*(i)} \right\rangle. \tag{6.6}$$

A direct algebraic manipulation of r_{k+1}^2 using the update $x_{k+1} = V_{i_k}(x_k)$ gives:

$$\mathbb{E}_{i_k}\left[\tfrac{1}{2} r_{k+1}^2 + f(x_{k+1}) - f^*\right] \;\le\; \tfrac{1}{2} r_k^2 + f(x_k) - f^* + \frac{1}{n}\langle \nabla f(x_k), x^* - x_k \rangle. \tag{6.7}$$

Here the expectation $\mathbb{E}_{i_k}$ is taking the random index i_k, conditional on x_k.

Since x^* is optimal and f is convex,

$$f(x^*) \ge f(x_k) + \langle \nabla f(x_k), x^* - x_k \rangle \quad \Rightarrow \quad \langle \nabla f(x_k), x^* - x_k \rangle \le f^* - f(x_k). \tag{6.8}$$

Plugging this into (6.7) yields

$$\mathbb{E}_{i_k}\left[\tfrac{1}{2} r_{k+1}^2 + f(x_{k+1}) - f^*\right] \;\le\; \tfrac{1}{2} r_k^2 + f(x_k) - f^* - \frac{1}{n}(f(x_k) - f^*). \tag{6.9}$$

Now take expectation over the whole history ξ_k and define $\phi_k = \mathbb{E}_{\xi_{k-1}} f(x_k)$ as before. Using $r_k^2 \le R_1^2(x_0)$ for all k, one obtains the recursive inequality

$$\phi_{k+1} - f^* \;\le\; \frac{n}{n+k+1}\left(\frac{1}{2} R_1^2(x_0) + f(x_0) - f^*\right). \tag{6.10}$$

Shifting the index from $k+1$ to k yields

$$\phi_k - f^* \;\le\; \frac{n}{n+k}\left(\frac{1}{2} R_1^2(x_0) + f(x_0) - f^*\right). \tag{6.11}$$

It shows an $O(\frac{n}{k})$ convergence rate for the expected objective function value in the constrained setting.

Strongly convex case

Assume now that f is strongly convex with respect to $\|\cdot\|_1$ with parameter $\sigma > 0$, gives

$$f(y) \ge f(x) + \langle \nabla f(x), y - x \rangle + \frac{\sigma}{2}\|y - x\|_1^2 \quad \forall x, y \in Q. \tag{6.12}$$

In particular, for $y = x^*$ and $x = x_k$ this gives

$$\langle \nabla f(x_k), x_k - x^* \rangle \;\ge\; f(x_k) - f^* + \frac{\sigma}{2} r_k^2 \;\ge\; \sigma r_k^2. \tag{6.13}$$

In terms of (6.7), this implies that the term involving $\langle \nabla f(x_k), x^* - x_k \rangle$ is negative and can be bounded more sharply. After some algebra, one obtains

$$\mathbb{E}_{i_k}\left[\tfrac{1}{2} r_{k+1}^2 + f(x_{k+1}) - f^*\right] \;\le\; \Big(1 - \frac{2\sigma}{n(1+\sigma)}\Big)\Big(\tfrac{1}{2} r_k^2 + f(x_k) - f^*\Big). \tag{6.14}$$

Taking total expectation over ξ_k and iterating this inequality yields

$$\phi_k - f^* \;\le\; \Big(1 - \frac{2\sigma}{n(1+\sigma)}\Big)^k \Big(\tfrac{1}{2} R_1^2(x_0) + f(x_0) - f^*\Big), \tag{6.15}$$

6.4 Comments

Compared with the unconstrained case in Section 4:

- The constrained step replaces the explicit gradient step $x - \frac{1}{L_i} U_i f_i'(x)^\#$ by a proximal projection onto Q_i, but the decrease estimate (6.5) has exactly the same structure.
- The sampling distribution is uniform ($p_i = 1/n$), so we no longer use the general $p_\alpha(i)$ from the unconstrained analysis.
- The convergence rate (6.11) is of order $O(n/k)$

Altogether, this Section shows that random coordinate descent extends naturally to constraints $Q = \bigotimes_i Q_i$, with essentially the same complexity dependence on n, the Lipschitz constants L_i, and the geometry of the initial level set.